

MetaGraphSci Architecture

Graph Transformer Self-Supervised Contrastive Neighbor Adaptation
for Scientific Document Classification

Project Milestone 2: Presentation of the First Version of the System

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1. Recap Problem Definition

Decision: Why target 1%, 5%, 10%, 25% labels?

Scientific domain expertise is scarce. High performance in low-label regimes is critical for specialized FoRC Field of Research Classification.

- MetaGraphSci treats citations as semantic proxies to bridge this label gap.
- **Goal:** Optimize a shared feature graph space where structural adjacency and textual semantics are co-aligned.

2. Recap Algorithm

Require: Graph $\mathcal{G}$, Documents $\mathcal{D} = \{\mathcal{D}_L, \mathcal{D}_U\}$, Modalities $\{\mathcal{T}, \mathcal{M}, \mathcal{N}\}$, Threshold β , Weight λ

1: $\theta, \{\mathbf{w}_c\} \leftarrow$ Initialize Transformer-QLoRA, Metadata DCN, Graph Transformer

2: **Stage 1: Meta-informed Contrastive Neighbor Adaptation MCNA**

3: **for** epoch $e = 1 \dots E_{pre}$ **do**

4: $\mathcal{B} \subset \mathcal{V} \leftarrow$ Sample mini-batch; $\mathbf{z}_i \leftarrow \Phi_\theta(\mathcal{T}_i, \mathcal{M}_i, \mathcal{N}_i)$

// Generate multimodal anchor

5: $\mathbf{z}_{i+} \leftarrow \Phi_\theta(\text{Augment}(\mathcal{T}_i), \mathcal{M}_i, \mathcal{N}_i)$

// Self-supervised positive view

6: $\mu_{ik} \leftarrow \text{Affinity}(\mathcal{M}_i, \mathcal{M}_k)$

// Identify metadata-compatible candidates

7: $\mathcal{L}_{NCNA} \leftarrow -\log \frac{\exp(\text{sim}(\mathbf{z}_i, \mathbf{z}_{i+})/\tau)}{\dots + \sum \omega_{ik} \exp(\text{sim}(\mathbf{z}_i, \mathbf{z}_k)/\tau)}$

// Mask $\mathcal{N}(i)$ to prevent false negatives

8: $\theta \leftarrow \theta - \eta \nabla_\theta \mathcal{L}_{NCNA}$

// Align representations with graph topology

9: **end for**

10: **Stage 2: Semi-Supervised Fine-Tuning SSFT**

11: **for** epoch $t = 1 \dots E_{fine}$ **do**

12: $p_L, p_U \leftarrow \Phi_\theta(\mathcal{B}_L, \mathcal{B}_U)$

// Compute posteriors for labeled and unlabeled streams

13: $\tilde{p}_U \leftarrow \text{Sharpen}(\text{Align}(p_U; \pi_{prior}))$

// Correct bias and minimize entropy

14: $T_c^{(t)} \leftarrow \beta \cdot \text{EMA}_c(t)$

// Update class-adaptive threshold via mastery

15: $M_u \leftarrow \mathbf{1}[\max \tilde{p}_u \geq T_{\hat{y}_u}^{(t)}]$

// Select high-confidence pseudo-labels $\hat{y}_u$

16: $\mathcal{L}_{total} \leftarrow \mathcal{L}_{sup}(\mathcal{B}_L) + \lambda \sum_u M_u \cdot \mathcal{L}_{ce}(\tilde{p}_u, \hat{y}_u)$

// Joint classification and consistency

17: $\theta, \{\mathbf{w}_c\} \leftarrow$ Update via $\nabla_{\theta, \mathbf{w}} \mathcal{L}_{total}$

// Optimize directional prototypes

18: **end for**

19: **return** $(\Phi_\theta, \{\mathbf{w}_c\}_{c=1}^C)$

3. Recap Evaluation Roadmap

Benchmarks & Scaling:

- **Standard Datasets:**
 - *OGBN-ArXiv*: 169k papers, large-scale benchmark.
 - *FoRC*: Multidisciplinary classification.
- **Data Scarcity**: Rigorous testing at 1%, 5%, and 10% label availability.
- **Statistical Variance**: Results reported as Mean $\pm$ Std Dev across 5 seeds.

Primary Classification Metrics:

- **Accuracy & Macro F1**: Standard and imbalance-aware performance.
- **AUROC & MCC**: Probability-aware and correlation-based reliability.
- **Calibration Analysis**: Reliability diagrams to detect overconfidence.

Scientific Visualization Library

Integrated pipeline for high-dimensional manifold analysis using **UMAP/t-SNE** and per-class error inspection using **Normalized Confusion Matrices**.

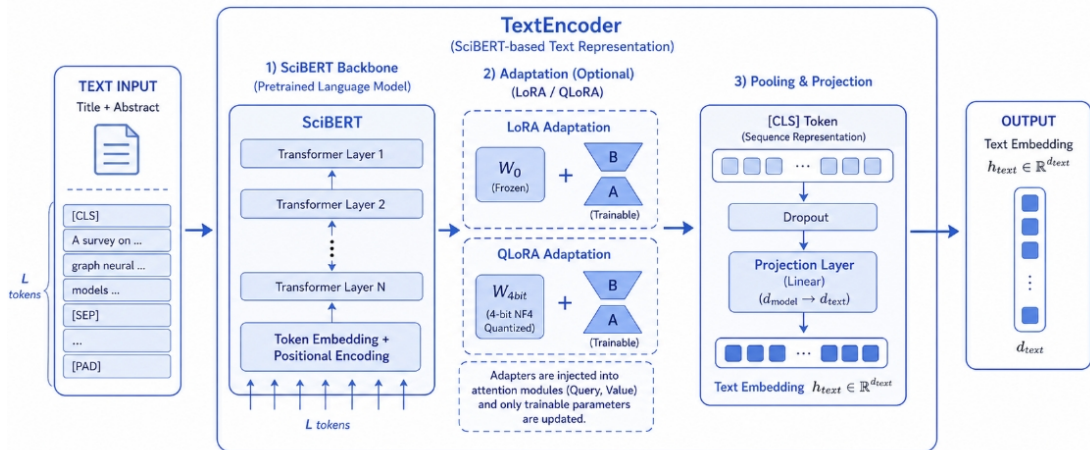
4. Architecture Overview

What we will cover in the software architecture

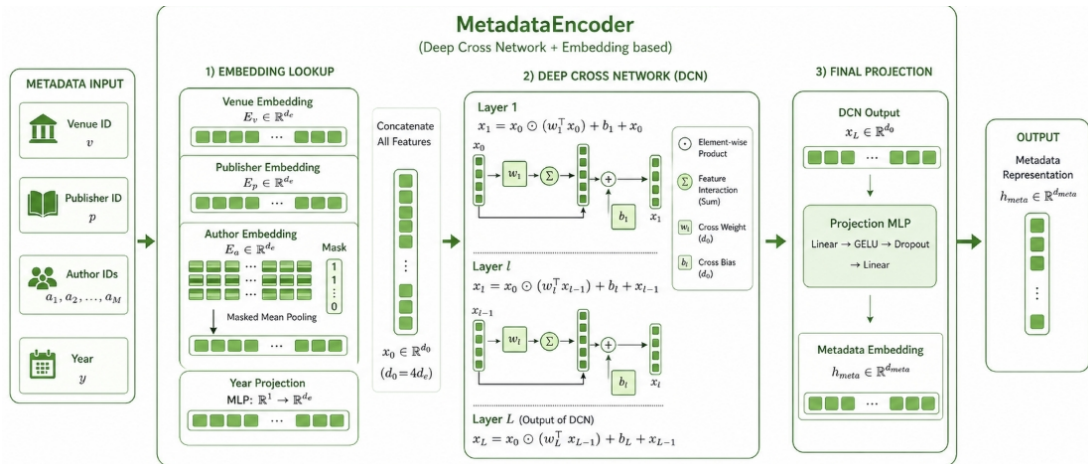
This is a roadmap of the core modules we will discuss, from input representations to training and execution flow.

- **Text Encoder:** builds semantic representations from scientific text.
- **Metadata Encoder:** processes structured attributes such as venue, authors, and auxiliary signals.
- **Multimodal Fusion:** combines text, metadata, and graph features into a shared representation.
- **Graph Transformer:** propagates context across citation and latent relations.
- **Training Strategy:** organizes pretraining, contrastive learning, and fine-tuning.
- **Workflow:** links ingestion, encoding, fusion, training, and evaluation in one pipeline.

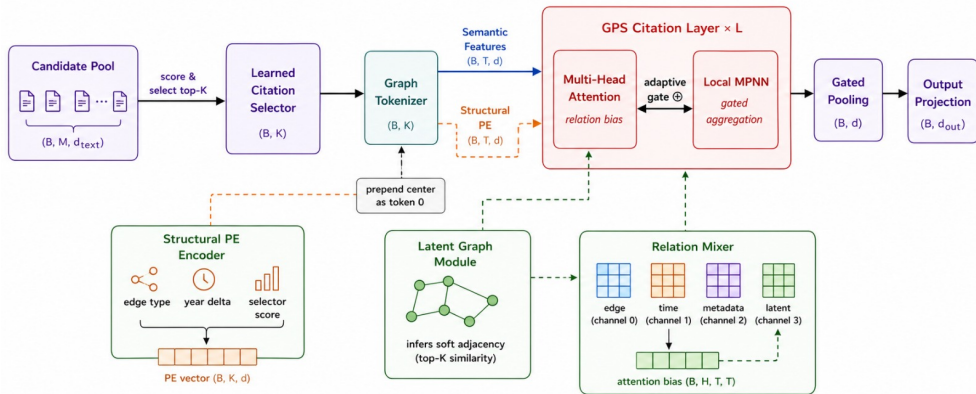
5. Text Encoder



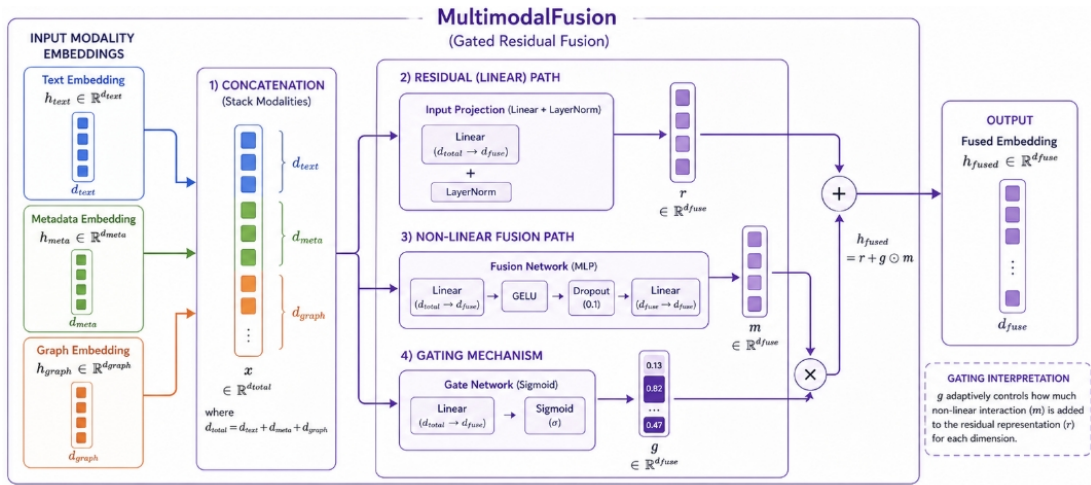
6. Metadata Encoder



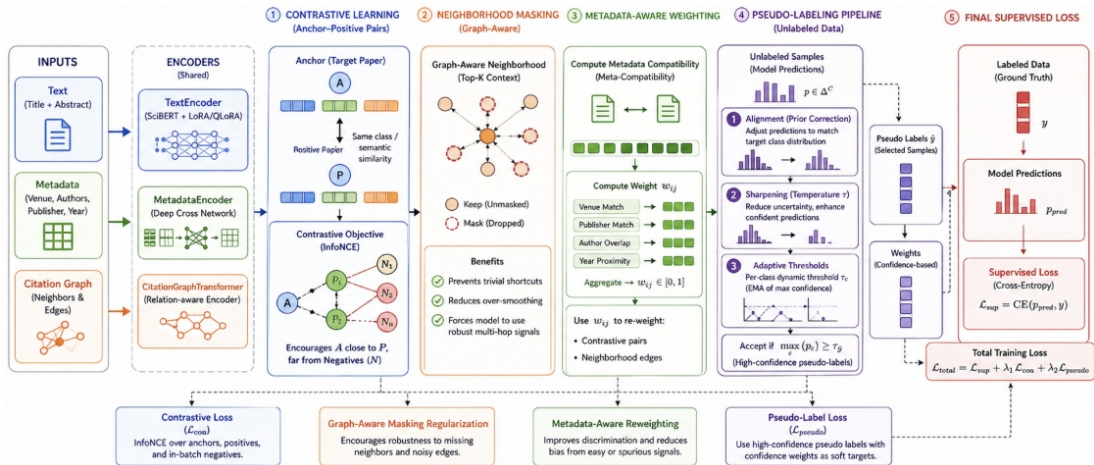
7. Graph Transformer



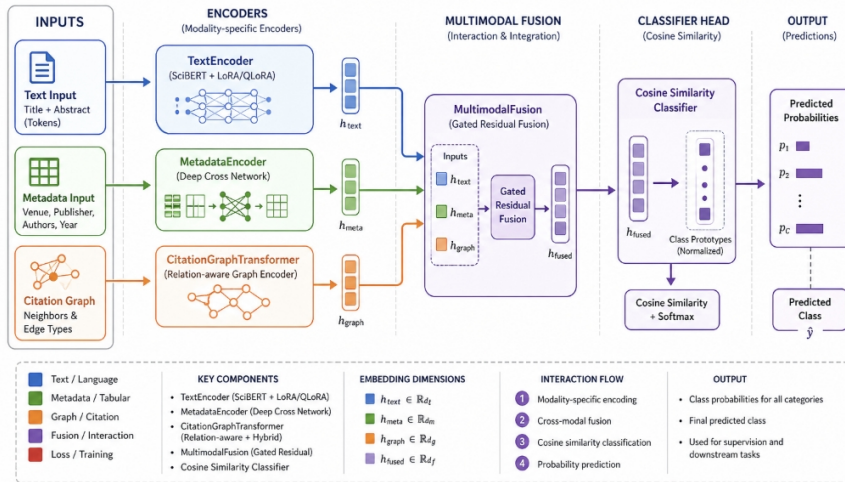
8. Multimodal Fusion



9. Training Strategy



10. Workflow



11. OpenAlex-AI Dataset Statistics

Dataset summary of scale, labels, metadata, text statistics, time span, and graph properties.

Corpus and Metadata

Characteristic	Value
<i>Volume</i>	
Number of documents	30 000
Number of citation edges	244 605
Nodes in citation graph	27 717
Isolated documents	2 283
<i>Labels</i>	
Number of classes	77
Labeling strategy	label_ratio = 0.10
Distribution	long-tail min 1, max 2 521
<i>Structured metadata</i>	
Unique venues	3 022
Unique publishers	606
Unique authors	60 417
Avg. authors per paper	3.65

Text, Time, and Graph

Characteristic	Value
<i>Text</i>	
Avg. title length words	8.2
Avg. abstract length words	146.5
Median abstract length words	141
<i>Temporal</i>	
Year range	1858–2025
Median / IQR year	2010 / [2003, 2018]
<i>Citation graph</i>	
Average out-degree	8.15
Evaluation type	transductive, temporal split

12. OpenAlex-AI Results

Classification Metrics				
Metric	full	text_only	text_metadata	text_citation
Accuracy	0.612 ± 0.008	0.548 ± 0.011	0.571 ± 0.009	0.594 ± 0.010
Balanced accuracy	0.341 ± 0.013	0.272 ± 0.015	0.298 ± 0.012	0.319 ± 0.014
Micro-F1	0.612 ± 0.008	0.548 ± 0.011	0.571 ± 0.009	0.594 ± 0.010
Macro-F1	0.378 ± 0.012	0.296 ± 0.014	0.327 ± 0.011	0.354 ± 0.013
Weighted-F1	0.583 ± 0.009	0.519 ± 0.012	0.544 ± 0.010	0.566 ± 0.011
Macro precision	0.402 ± 0.014	0.318 ± 0.016	0.349 ± 0.013	0.378 ± 0.014
Macro recall	0.367 ± 0.011	0.288 ± 0.013	0.316 ± 0.011	0.343 ± 0.012
Matthews CC	0.581 ± 0.010	0.512 ± 0.013	0.539 ± 0.011	0.563 ± 0.012
Cohen's κ	0.578 ± 0.010	0.508 ± 0.013	0.535 ± 0.011	0.560 ± 0.012
Macro OvR AUROC	0.812 ± 0.006	0.764 ± 0.009	0.781 ± 0.007	0.799 ± 0.007
Log-loss	1.842 ± 0.031	2.118 ± 0.042	2.014 ± 0.037	1.913 ± 0.034

Efficiency				
Metric	full	text	text+meta	text+cit
Time / epoch min	~18	~11	~13	~16
Peak VRAM GB	~22	~15	~17	~20